

SEMAI Forecast: references

Brand New Developers · PSC GET Talent 2026 · Machine Learning (software)

Code, dataset and trained model: github.com/YochananAodh/semai-ml (notebook

SEMAI_Forecast.ipynb)

Academic sources (APA 7)

1. Allen, R. G., Pereira, L. S., Raes, D., & Smith, M. (1998). *Crop evapotranspiration: Guidelines for computing crop water requirements* (FAO Irrigation and Drainage Paper No. 56). Food and Agriculture Organization of the United Nations.

Used for: how solar radiation over the day drives evaporation and heating (top feature #2, hour of day), and the FAO-56 reference evapotranspiration (ET_0) column in the dataset.

2. Grossiord, C., Buckley, T. N., Cernusak, L. A., Novick, K. A., Poulter, B., Siegwolf, R. T. W., Sperry, J. S., & McDowell, N. G. (2020). Plant responses to rising vapor pressure deficit. *New Phytologist*, 226(6), 1550–1566. <https://doi.org/10.1111/nph.16485>

Used for: why VPD, not humidity alone, drives plant water stress (the misting label and top feature #1, VPD now).

3. Hersbach, H., Bell, B., Berrisford, P., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730), 1999–2049. <https://doi.org/10.1002/qj.3803>

Used for: the reanalysis data that Open-Meteo's historical archive is built on (data quality and limits).

4. Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Used for: the logistic regression and gradient-boosting models, permutation importance and all metrics.

5. Seneviratne, S. I., Corti, T., Davin, E. L., Hirschi, M., Jaeger, E. B., Lehner, I., Orlowsky, B., & Teuling, A. J. (2010). Investigating soil moisture–climate interactions in a changing climate: A review. *Earth-Science Reviews*, 99(3–4), 125–161. <https://doi.org/10.1016/j.earscirev.2010.02.004>

Used for: why drier topsoil means less evaporative cooling and faster-rising VPD (top feature #3, soil moisture).

6. Shamshiri, R. R., Jones, J. W., Thorp, K. R., Ahmad, D., Man, H. C., & Taheri, S. (2018). Review of optimum temperature, humidity, and vapour pressure deficit for microclimate evaluation and control in greenhouse cultivation of tomato: A review. *International Agrophysics*, 32(2), 287–302. <https://doi.org/10.1515/intag-2017-0005>

Used for: the greenhouse VPD range that sets the rig's 1.2 kPa misting threshold.

7. Zippenfenig, P. (2023). *Open-Meteo.com Weather API* [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.7970649>

Used for: the dataset itself (see below).

Dataset

Source	Open-Meteo Historical Weather API (ERA5 reanalysis), downloaded as CSV
Location and period	Penang, 5.4482° N, 100.29° E · hourly, 1 Jan 2023 – 31 Dec 2025 · local time (Asia/Kuala_Lumpur)
Size	Raw: 26,304 rows × 26 columns (25 weather variables + time). Cleaned: 26,301 rows × 31 columns.
Licence	Creative Commons Attribution 4.0 (CC BY 4.0). Weather data by Open-Meteo.com. Real data only, nothing synthetic.
Files in the repository	dataset/penang_weather_hourly_2023_2025_raw.csv · dataset/penang_weather_hourly_2023_2025_clean.csv

https://archive-api.open-meteo.com/v1/archive?latitude=5.4482&longitude=100.29&start_date=2023-01-01&end_date=2025-12-31&hourly=temperature_2m,relative_humidity_2m,dew_point_2m,apparent_temperature,precipitation,rain,cloud_cover,cloud_cover_low,cloud_cover_mid,cloud_cover_high,shortwave_radiation,direct_radiation,diffuse_radiation,sunshine_duration,wind_speed_10m,wind_direction_10m,wind_gusts_10m,surface_pressure,power_msl,et0_fao_evapotranspiration,vapour_pressure_deficit,soil_temperature_0_to_7cm,soil_moisture_0_to_7cm,is_day,weather_code&timezone=Asia%2FKuala_Lumpur&format=csv

Software

Python 3.12 · pandas 3.0 · NumPy 2.5 · scikit-learn 1.9 · Matplotlib 3.11 · Jupyter. All open source.

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